

Community resilience to wildfires: A network analysis approach by utilizing human mobility data

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ABSTRACT

Disasters have been a long-standing concern to societies at large. With growing attention being paid to resilient communities, such concern has been brought to the forefront of resilience studies. However, there is a wide variety of definitions with respect to resilience, and a precise definition has yet to emerge. Moreover, much work to date has often focused only on the immediate response to an event, thus investigating the resilience of an area over a prolonged period of time has remained largely unexplored. To overcome these issues, we propose a novel framework utilizing network analysis and concepts from disaster science (e.g., the resilience triangle) to quantify the long-term impacts of wildfires. Taking the Mendocino Complex and Camp wildfires - the largest and most deadly wildfires in California to date, respectively - as case studies, we capture the robustness and vulnerability of communities based on human mobility data from 2018 to 2019. The results show that demographic and socioeconomic characteristics alone only partially capture community resilience, however, by leveraging human mobility data and network analysis techniques, we can enhance our understanding of resilience over space and time, providing a new lens to study disasters and their long-term impacts on society.

1. Introduction

Disasters, such as floods, earthquakes and wildfires have been a long-standing concern to societies across the globe. They often result in unpredictable effects on the environment, wildlife as well as human population. For example, disasters can disrupt transportation networks, supply chains, cause property loss, and can impact the biodiversity of a region. While taking wildfires as a disaster example, it is evident that, over recent decades, the frequency and impact of these events have not only increased in the United States but also in various parts of the world (e.g., Abatzoglou & Williams, 2016; Higuera, Abatzoglou, Littell, & Morgan, 2015; Running, 2006; USGCRP, 2018; Westerling, 2016). This trend perhaps best exemplified in countries like Greece and Spain, where multiple devastating wildfires have occurred in recent years, highlighting the global nature of this issue. The increasing occurrences of wildfires worldwide, driven by factors such as climate change and population growth, underscores the urgent need for a comprehensive understanding of community resilience to such extreme events. However, the exact definition of resilience has yet come to the consensus as it relates to a diverse set of determinants, ranging from individuals, society, to that of the economy and environment which all come into play

when a disaster strikes (McCubbin, 2001; Southwick, Bonanno, Masten, Panter-Brick, & Yehuda, 2014; Yabe, Jones, Rao, Gonzalez, & Ukkusuri, 2022). That being said, in several disaster resiliency studies, one major notion of resilience stresses how well a system can re-establish stability and functions from the disruption of a disaster over time. For example, the National Academies (2012) defines disaster resilience as “the ability to prepare and plan for, absorb, recover from, or more successfully adapt to actual or potential adverse events.”

In this context, questions regarding how we quantify resilience to disasters have drawn attention from researchers across many disciplines. One commonly used approach is developing indices derived from different resilience determinants to evaluate and compare disaster resilience, such as the Baseline Resilience Index for Communities (BRIC) (Cutter, Burton, & Emrich, 2010), Community Disaster Resilience Index (CDRI) (Peacock et al., 2010), and Resilience Capacity Index (RCI) (Foster, 2012). These indices, and others alike, aggregate different metrics (e.g., social, economic, infrastructure, environment) to create a comparative resilience score, allowing the comparison of disaster resilience across space. However, their disadvantage relates to their performance in evaluating resilience which varies from one to the other, making the selection of the appropriate index to assist in emergency

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management and policy decision-making extremely difficult (Bakken, Fox-Lent, Read, & Linkov, 2017). Besides, as discussed above, each index is derived from multiple aggregated metrics, which makes the contribution of each metric unclear (Bakkensen et al., 2017). Moreover, such indices generally view resilience statically as an outcome for a specific moment in time. In other words, they overlook the dynamic changes of resilience over time (Southwick et al., 2014).

Another major focus of disaster resilience research is the study of human behaviors. Such studies aim to understand how people behave and interact with the built and natural environment in times of disasters from both individual and collective aspects. For example, in terms of collective behaviors, studies have uncovered the role of social relationships in supporting and adaptation in disasters (e.g., Aguirre, Wenger, & Vigo, 1998; Bolin, 2007; Drabek, Key, Erickson, & Crowe, 1975; Quarantelli, 1984). Regarding individual behaviors, research generally focuses on cognitive science, ranging from understanding how people react to stresses in disasters (e.g., Fritz & Marks, 1954; Glass, 1959; Quarantelli, 1954), to social/place attachments (e.g., Grinberger & Felsenstein, 2016; Hobfoll, 1989; Hobfoll, 2001; Mawson, 2005) and decision-making (e.g., Bandura, 2001; Benight & Bandura, 2004; Paton, 2003; Richard Eiser et al., 2012; Rosenstein, 2004). Although these studies provide more dimensions, such as perspectives of social norms, psychology, and social attachment, in disaster resilience research, they have mostly focused on the theoretical construction and/or conceptual modeling (Burger, Kennedy, & Crooks, 2021), where empirical validation still largely remains unexplored due to, for instance, the limitation of available data.

The advent of Web 2.0 and associated technologies (O'Reilly, 2006) has led to the emergence of “big data” which has proliferated due to location-based services, social media platforms (e.g., Twitter, Facebook) and mobile phone devices. They have produced data at an unprecedented scale, resulting in disaster resilience research shifting towards a more data-driven direction (Burger, Oz, Kennedy, & Crooks, 2019; Hsu, Liu, Nguyen, Chien, & Mostafavi, 2024). One major stream of this data-driven approach is the use of sentiment analysis techniques to investigate people's opinions to disasters. For example, Yuan, Li, Liu, Zhai, and Qi (2021) utilized Twitter data to identify dynamic crisis response disparities and explore concern themes of individuals with different demographic characteristics (e.g., race, gender) during Hurricane Florence. Similarly, Yan, Chen, and Wang (2020) used geotagged Twitter data to identify and compare people's sentiments and perspectives of post-disaster recovery process by using sentiment analysis and latent dirichlet allocation topic modeling after earthquakes in Lombok and Bali. Another stream focuses on human mobility analysis to monitor and compare individuals' movements and behavioral responses to a disaster. For instance, Roy, Cebrian, and Hasan (2019) introduced the notion of mobility resilience to detect the ability of people to resist impacts of an extreme event (e.g., hurricane, earthquake, rain storm) by using geo-located social media data. Yabe, Zhang, and Ukkusuri (2020) studied business recovery in Puerto Rico after Hurricane Maria based on mobile phone data provided by Safegraph. Through the analysis of daily visits of establishments, they were able to quantify the disaster impacts to different types of businesses.

The above studies along with several others have illustrated the potential and flexibility of using big data to connect disaster information and people in times of crises, enhancing the understanding of disaster resilience. However, many studies have often focused on the short-term analysis (i.e., the response), which only covers one of the four stages of disaster management (i.e., mitigation, preparedness, response, and recovery). One reason for this is that analyzing resilience over a long-term period is still a challenge. Against this backdrop, we propose a novel framework that incorporates network analysis and the concept of the resilience triangle to capture potential impacts of dynamic disruptions of a disaster, especially on collective human behaviors to assess a community's resilience to wildfires. Specifically, we employed degree centrality, a key metric in the network analysis, as a proxy for evaluating

disaster resilience. The degree centrality reflects the extent to which a community-based group has been exposed to a disaster, where a higher degree centrality implies greater exposure. These measures have been applied in urban resilience studies, especially relating to topics around natural disasters, climate change, transport, urban forms and social ties (e.g., Akbarzadeh, Memarmontazerin, Derrible, & Salehi Reihani, 2019; Boulmakoul, Badaoui, Karim, Lbath, & Oulad Haj Thami, 2022; Kalpana, Jayasinghe, Abenayake, & Wijayawardana, 2021; Torres & Casey, 2017; Wang, Chen, Mu, & Zhang, 2020). By analyzing the dynamic changes in degree centrality of a community-based group, we are able to assess and quantify resilience in the face of wildfire disruptions in this paper.

Complementing our network analysis, we utilized the concept of the resilience triangle, which originated from the work of Bruneau et al. (2003), as a quantitative conceptual framework for measuring disaster resilience as shown in Fig. 1. The concept was then mathematically modeled by Tierney and Bruneau (2007) and Cimellaro, Reinhorn, and Bruneau (2010). In general, the resilience triangle can be used to measure the abrupt functionality loss of a social unit (e.g., a community) over time under the disruption of a disaster. The depth of the triangle (highlighted with red dashed line in Fig. 1) illustrates the severity of the disruption, the length of it demonstrates the recovery time, and the area of it indicates the resilience of the social unit. Fig. 1 (a) schematically represents the resilience triangle, where the y-axis depicts the quality of infrastructure $Q(t)$ as a function of time, t_0 marks the occurrence of a disruptive event, and t_1 indicates the time when the infrastructure has completely recovered. In our study, we take into account the deteriorating situation as noted in the work of Hossain, Roy, Mohammad, Nawar, and Dipta (2021). Our rationale being that several studies have shown that the deteriorating situation is not an abrupt change but gradual drop after t_0 (e.g., Roy et al., 2019), which is shown in Fig. 1 (b), specifically the path between A to B, where B is the lowest value at time t_0 - a point defined as “systemic impact” by Vugrin, Warren, Ehlen, and Camphouse (2010). This point represents the difference between the intended and actual performance levels following a disruption. Building upon the resilience triangle, resilience has been formally quantified as the ratio of the area of the triangle (the region shaded orange in Fig. 1 (b)) to the total area under the pre-disruption level of performance, from the initial disturbance at t_0 to the recovery at t_1 (i.e., the sum of the areas shaded orange and green in Fig. 1 (b)) (Attoh-Okine, Cooper, & Mensah, 2009). A smaller ratio indicates a higher level of resilience (Labib, 2021).

In addition, the resilience triangle also allows us to capture two important features of resilience, namely robustness (referring to the ability of a social unit to cope with the impacts of disruption after a disaster) and vulnerability (which refers to the potential for loss (Cutter, Boruff, & Shirley, 2003)) of a social unit (see Fig. 1(b)). This triangle concept has been widely applied in different domains, such natural disasters (e.g., Hossain et al., 2021; Kahnemouei, Bolandi, & Haghifam, 2017; Mudigonda, Ozbay, & Bartin, 2019; Zobel & Khansa, 2014), supply chains (e.g., Adams, Bekkem, & Toledo-Dur'an, 2012; Bevilacqua, Ciarapica, & Marcucci, 2018; Pant, Barker, Ramirez-Marquez, & Rocco, 2014), civil infrastructure as well as economic systems (e.g., Yang & Frangopol, 2019), indicating the broad acknowledgement of the applicability of using the concept for quantifying resilience.

In this study, we evaluate the community resilience by combining network analysis, dynamic time warping clustering, as well as regression analysis techniques with the concept of the resilience triangle in space and time. Doing this allows us to answer questions, such as which community is more resilient compared to others and why? Did a community bounce back to its original status after a certain time or form a new normalcy under the effects of disruptions? What are the similarities and differences among communities? Answering these questions can help understand the resilience of each community, as well as the potential linkages between communities. Moreover, we argue that the proposed method can help scale up the concept of resilience to a more

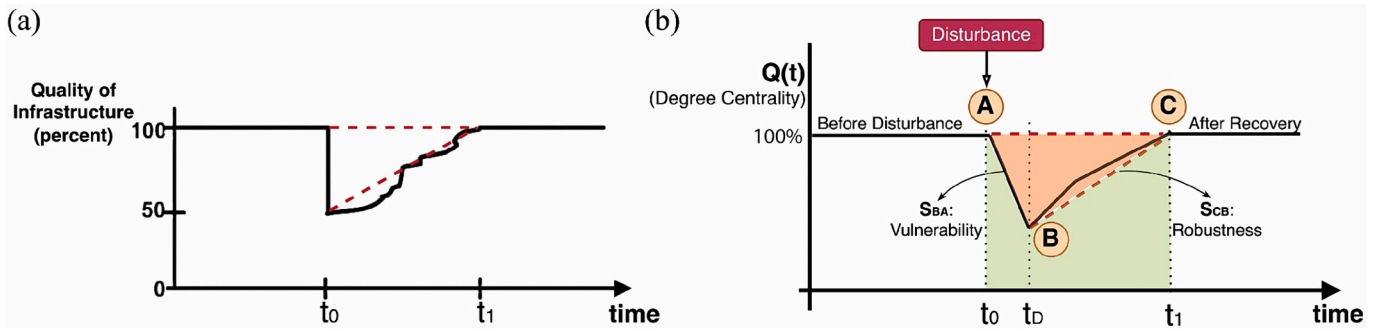


Fig. 1. Resilience triangle. (a) The original resilience triangle (adapted from Bruneau et al., 2003); (b) The modified resilience triangle used in this study.

empirical framework that can be quantified and visualized through computational techniques assistance and data. In what follows, we present our methodology (Section 2) and then results and discussion (Section 3). This is followed in Section 4 brief conclusion and areas of further work.

2. Methodology

In this paper, we take the Mendocino Complex (MC) and Camp wildfires - the largest and most deadly wildfires in California to date, separately, as case studies to evaluate community resilience from both spatial and temporal perspectives (Fig. 3(a)). More precisely, we first introduce the study area (Section 2.1) and mobility data utilized (Section 2.2). Building upon what we discussed in introduction (Section 1), we then analyze the degree centrality derived from the human mobility data (Section 2.3), which is then used in the subsequent resilience triangle detection and Dynamic Time Warping (DTW) clustering (Section 2.4 & 2.5). Lastly, in Section 2.6, we explore the underlying connections between community resilience and related demographic characteristics, socio-economic status and built environment characteristics via

regression analysis. Fig. 2 illustrates an overview of the research outline.

2.1. Study area

As noted in the introduction, Wildfires have been a long-standing concern to societies at large. According to the official United States wildfire records provided by the National Interagency Fire Center (NIFC), the total number of wildfires has declined over the past decade as shown in Fig. 3 (c). However, there has been a considerable increase in the acres burned in recent years, indicating that wildfires are much more destructive and deadly than ever before. California - the most populous state in the United States (Census Bureau Data, 2023) - ranks as the most wildfire-prone state in the country (Verisk, 2021), putting it at the greatest risk of wildfire damage as can be seen in Fig. 3 (b). In this context, understanding the impacts of wildfires and the resiliency of areas that are often exposed to such natural disasters become essential. Table 1 displays the basic information about the two selected wildfires - MC and Camp - in this study.

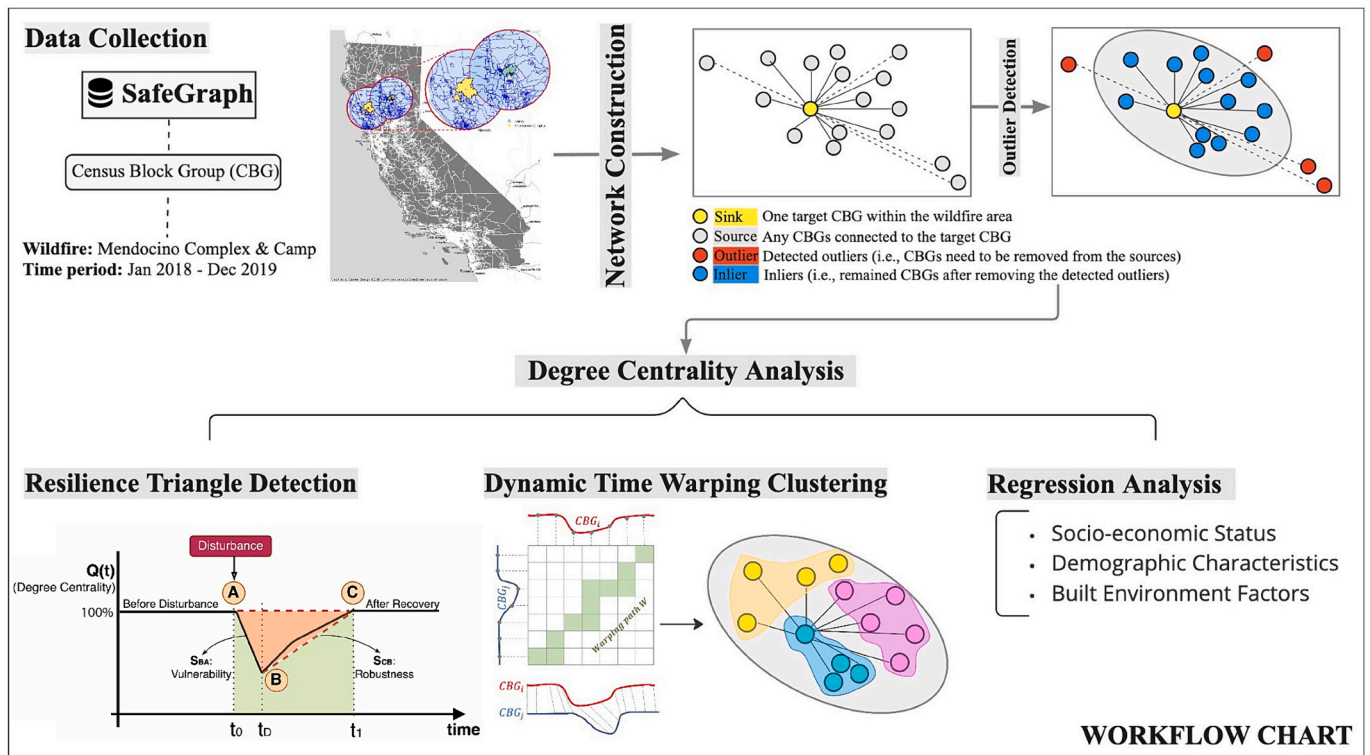


Fig. 2. An overview of the research outline.

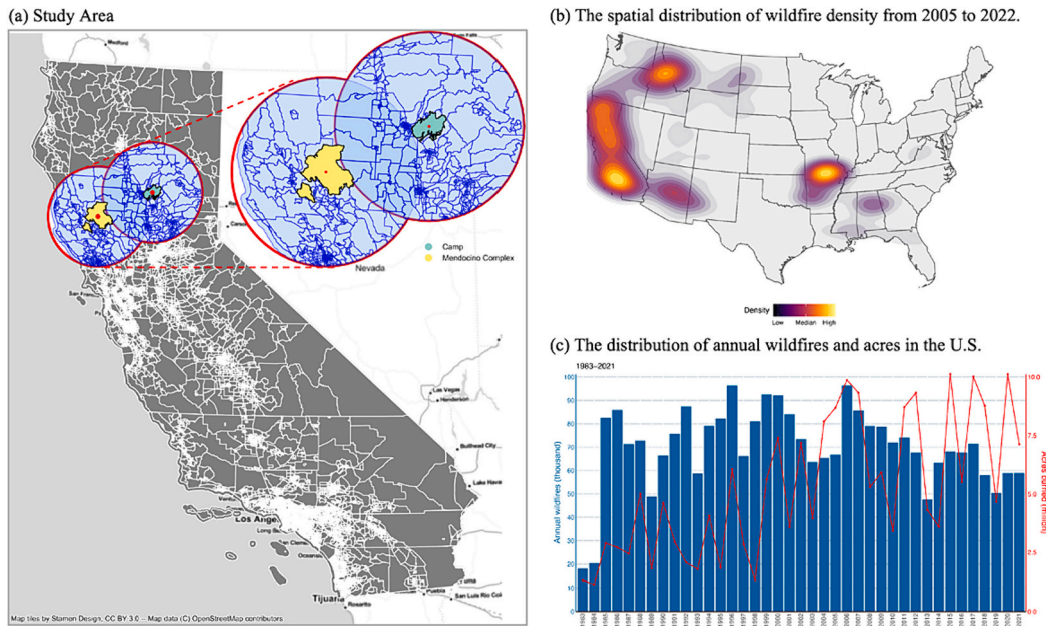


Fig. 3. (a) The zoomed in study areas of the two wildfires, where the blue areas highlight the Census Block Groups; (b) The spatial distribution of wildfire density from 2005 to 2022; (c) The distribution of annual wildfires and acres in the U.S. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Description of wildfires focused on the study (Sources: National Centers for Environmental Information).

Wildfire name	Start Date	End Date	Description
Mendocino complex (MC)	08/27/2018	01/04/2019	The largest wildfire to date with records back to 1933, which has burned over 450 k acres.
Camp	11/8/2018	11/25/2018	The costliest and deadliest wildfire, which has destroyed >18,500 buildings.

2.2. Data collection

As mentioned in Section 1, this study proposes an ensemble approach that combines network analysis and concepts from disaster science (i.e., the resilience triangle) to quantify the long-term impacts of wildfires, especially on collective human behavior. To illustrate this potential, we utilize a human mobility data collected from SafeGraph – a commercial provider of geospatial data in the United States (SafeGraph, 2023). SafeGraph made their data publicly available to the research community during the COVID-19 pandemic. This dataset includes patterns data, which are aggregated at the Census Block Group (CBG) level, providing information about the monthly frequency and destinations of visits. This granularity enables us to observe mobility trends between communities, with CBGs typically consisting of 600 to 3000 individuals. Such details allow for a granular view of community dynamics, which is critical for evaluating the impact of wildfires on human movement patterns. To illustrate the dataset's form and the type of information it encompasses, Table A1 in Appendix provides examples of the attributes included in the data. It should be noted that the data is not without its sources of uncertainty, such as bias at an aggregation level and in demographics (SafeGraph, 2019). Yet, to develop a more comprehensive understanding of such uncertainties is beyond the scope of the current paper. We will come back to this issue in Conclusion (Section 4). Nonetheless, the wide coverage of the data across the United States makes it a valuable source for capturing mobility patterns, which has been shown in many recent studies (e.g., Chang et al., 2021; Hu, Quigley, & Taylor, 2021; Yabe et al., 2022).

Our focus on this data spanning from January 2018 to December 2019. This period is selected primarily due to the availability of consistent data from SafeGraph, but at the same time to avoid potential biases caused by the COVID-19 pandemic, which significantly altered human mobility patterns (Li, Zhao, Haitao, Mansourian, & Axhausen, 2021; Zhang et al., 2022). While this two-year period may seem limited, it coincides with the median recovery timeline for wildfires (Crutchfield, 2013), thereby allowing us to capture insights into community responses and recovery processes from such extreme events during this critical phase. We extract the visitation frequency data for each CBG (i.e., from the home CBG to any other CBGs) on a monthly basis, which is the finest time interval in the dataset. This long-term mobility data is particularly crucial for our study, aligning with the resilience triangle framework to examine the prolonged process of community adaptation and recovery over time. Observing these mobility patterns on a monthly basis allows us to uncover how communities navigate through different phases (i.e., response and recovery) following wildfire events. The data is then filtered to focus on CBGs impacted by the wildfires, thus enabling us to assess changes in human mobility patterns and infer aspects of community resilience during this period.

2.3. Centrality measurement

Once data has been collected, we then turn our attention to centrality measurement. Centrality measures are commonly used for evaluating the degree of importance of specific nodes or links in a network. In this work, we take the degree centrality as a proxy for disaster resilience evaluation, which demonstrates the number of links that involve a given node (i.e., a CBG). Specifically, we take each CBG impacted by the wildfire as a “destination” node and delineate the movement of individuals from their “home” CBG to these “destination” CBGs as links. These links are weighted by the aggregated frequency of visitation during a given month as derived from the SafeGraph data, which reflects the strength of interaction between CBGs in response to wildfire events. In doing so, we can construct a network that captures all pairs of home-to-destination CBGs, serving as the basis for our subsequent degree centrality analysis.

2.3.1. Outlier detection

Before operationalizing centrality measures (which we outline in Section 2.3.2), outlier detection is conducted to eliminate potential impacts on network measures caused by extreme node(s). Specifically, an outlier node here is considered as a source node (i.e., a CBG with visitors to a target CBG within the wildfire area) with an unusual distance to the target node (i.e., the target CBG) compared to that of all the other source nodes, which can be sensitive in the network analysis. Against this backdrop, we do so by removing anomalous observations from data to improve the robustness of the network.

To conduct outlier detection, Mahalanobis Distance (MD: Mahalanobis, 1936) is used as the evaluation metric for identifying outlier nodes (e.g., CBGs). MD is a measure of distance between a point and a distribution, based on a covariance matrix between multivariate data. Unlike the commonly used Euclidean distance that measures the length between two points in Euclidean space, MD calculates the distance between two points based on the distribution pattern of data points themselves. In other words, it considers how many standard deviations away a point is from the center of a distribution and it has been shown to work especially well for two or more variables that are highly correlated (Ghorbani, 2019).

In our analysis, the specific variables used for MD calculations are the geographical coordinates of the centroids of the CBGs. These geographical coordinates are inherently correlated, which makes them particularly suitable for MD analysis (Mahalanobis, 1936). For each CBG within the wildfire area, identified as the target CBG node, we examine the distribution of all other CBGs connected to it based on their centroid locations. The MDs are then calculated between the target CBG node and its connected CBGs using Eq. 1, where X_{p1} refers to the coordinates of the target CBG node, X_{p2} refers to the coordinates of any other CBG node that is connected to it, and C^{-1} represents the negative first power of the covariance matrix. This method is applied to each considered CBG node within the wildfire area, enabling us to identify outlier nodes based on their spatial relationships effectively.

$$D^2 = (X_{p1} - X_{p2})^T C^{-1} (X_{p1} - X_{p2}) \quad (1)$$

Fig. 4 gives an example of the detected outliers based on the method discussed above with different confidence intervals using Chi-Square

values. Specifically, Fig. 4 (b) demonstrates a simplified network where the yellow center point represents the target CBG, and the gray points refers to its connected CBGs. The MD aforementioned is applied to measure the distance between the yellow point to each of the gray points for identifying outliers. To visually observe the detected outliers, an ellipse is drawn based on the covariance matrix of the points, where the radius is defined according to the Chi-Square distribution. As such, points outside the ellipse (red crosses) are considered as the outliers and points inside the ellipse (blue circles), named inliers, are retained. In addition, different radii are defined from the Chi-Square distribution (ci equal to 80%, 90% and 98% separately) to examine the sensitivity of Chi-Square selection for removing outliers on the degree centrality measures. The results show that the choice of Chi-Square does not affect the degree centrality measures, as such, we use 98% probability in the Chi-Square distribution to define the radius of the ellipse. The reason for choosing this value is to keep as many of the points as possible but at the same time, to remove anomalous points from the distribution to improve the robustness of the network.

2.3.2. Degree centrality analysis

Once outlier detection has been carried out, we can then move towards measuring degree centrality. Degree centrality is an index of exposure to what is flowing through the network. This can be interpreted as an opportunity to influence and to be influenced directly by a specific flow. Linking this concept to disaster resilience and to this paper, a node or link with high degree centrality indicates higher probability to be disrupted when being hit by a disaster, affecting its reliability and resilience (Sharifi, 2019). In this regard, we take degree centrality, or more specifically, in-degree centrality, as a proxy for disaster resilience to explore how wildfires in California affect the inflow of mobility between CBGs in a long-term period. We chose in-degree centrality as we are interested in how the visiting patterns of an area changed under the impact of a wildfire. This allows us to question if the degree centrality of CBGs decrease after the wildfires. Specifically, this allows us to explore the following questions: when does it reach its lowest value and when does it starts to increase (i.e., the stage of recovery)? Furthermore, can the degree centrality return to its original status or establish a new normalcy after the impact of a wildfire? And, are there any similarity between different CBGs?

(a) An example of detected outliers

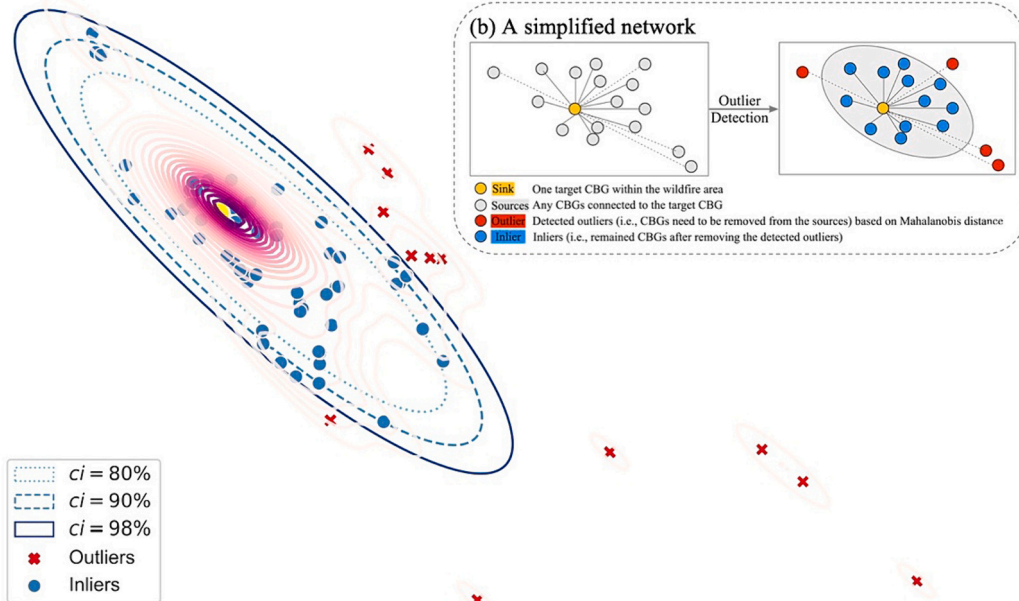


Fig. 4. An example of outlier detection with different Chi-Square values.

To answer these questions, we measure the in-degree centrality weighted by visitation frequency for each CBG from January 2018 to December 2019 in monthly intervals. Furthermore, in order to make it easier to compare different wildfire events, we normalize the in-degree centrality outcomes by using the Min-Max normalization. The results are then used in the subsequent resilience triangle detection (Section 2.4) and CBG clustering (Section 2.5).

2.4. Resilience triangle detection

As discussed in Section 1, we are using the concept of the resilience triangle to quantify resilience. In order to achieve this, resilience triangles are constructed based on the degree centrality curves introduced in Section 2.3. To help introduce the resilience triangle detection procedure, we will now present two examples of detected triangles as shown in Fig. 5. As inputs to the detection algorithm, the blue curves indicate the average degree centrality of the CBGs within the Camp and MC wildfires, separately. The dashed green line represents the pre-wildfire baseline level (i.e., normal status), which is defined as the two-month average degree centrality right before a disaster happens. Following the definition in Bruneau et al. (2003), the first vertex of triangle is defined by the onset of a disaster, denoted as t_0 . The second vertex t_D , is the time when the degree centrality reaches its minimum value between t_0 and t_1 . The third vertex, t_1 , refers to the time when the degree centrality fully bounces back to the pre-wildfire baseline level (see Fig. 5, right). However, if a CBG never fully restores to the baseline level (i.e., forms a new normalcy), in this case, t_1 is assigned by the time when the maximum degree centrality occurs after the wildfire becomes inactive (see Fig. 5, left).

The identified resilience triangle for each CBG enables us to further derive two resilience measures, namely vulnerability and robustness. These are measured based on the slopes between t_0 and t_D for vulnerability and for robustness, the slope between t_1 and t_D using Eq. 2 & Eq. 3, where p refers to a CBG, and $deg.$ refers to the degree centrality. This approach follows the framework provided by Serdar, Koç, and Al-Ghamdi (2021).

$$vulnerability = \frac{deg_{t_D}(p) - deg_{t_0}(p)}{t_D - t_0} \quad (2)$$

$$robustness = \frac{deg_{t_1}(p) - deg_{t_D}(p)}{t_1 - t_D} \quad (3)$$

2.5. Dynamic time warping clustering

In order to compare similarity between the different CBGs, we use

the Dynamic Time Warping (DTW) clustering technique (Berndt & Clifford, 1994) to conduct the similarity measure, which has been noted as being an accurate method for clustering time series data (Wang et al., 2013). It is based on the concept of a warping path, which identifies the optimum matching (i.e., to minimize distance) between two sequences using a dynamic programming approach (Wiki, 2023; Zhang et al., 2017). In the context of this study, each CBG can have a different response and recovery pattern when being impacted by a wildfire due to its level of resilience. In other words, every CBG has its own changing pattern of degree centrality from the time before the onset of a wildfire to the time of its recovery. That being said, we can classify CBGs based on their changing patterns of degree centrality to evaluate the similarity between them. To operationalize the similarity measure using DTW, we consider each CBG as a univariate time series (i.e., an observation), where each column is the degree centrality at a specific time point (i.e., a month in our case). As such, we create two time series datasets for the two wildfire events separately. For the MC wildfire, due to it containing 11 CBGs, we create a set of 11 time series. Similarly, for the Camp wildfire, we create a set of 42 time series. Appendix B1 gives an example of how the warping path has been identified for two degree centrality sequences of two examples of CBGs. Afterwards, the measured DTW distances are used as the basis for grouping CBGs into clusters, a process achieved by leveraging Hierarchical Agglomerative Clustering (HAC) (Dash, Liu, Scheuermann, & Tan, 2003). HAC generates a dendrogram – a tree-like diagram that visualizes the arrangement of clusters formed based on the similarity within the data – allowing us to identify and interpret clusters of CBGs with similar degree centrality patterns over time.

2.6. Regression analysis

The final step of our methodology is to contextualize the resilience with respect to the disruption of wildfires. Specifically, we investigate the degree centrality metric (introduced in Section 2.3.2) with the demographic, socio-economic as well as built environment characteristics of each CBG through regression analysis. We do so by creating 9 independent variables as shown in Table 2 from publicly available datasets (e.g., Census data) at the CBG level (i.e., with the same spatial resolution as that in degree centrality analysis), which are often encountered in existing literature (e.g., Bakkenen et al., 2017; Frazier, Thompson, Dezzani, & Buttsick, 2013). It is necessary to note that the independent variables related to demographic and socio-economic status may not change as much as the mobility data, but it is crucial to point out that the objective for conducting the regression analysis does not intend to predict degree centrality based on these independent variables but instead to provide an initial quantitative exploration of the potential

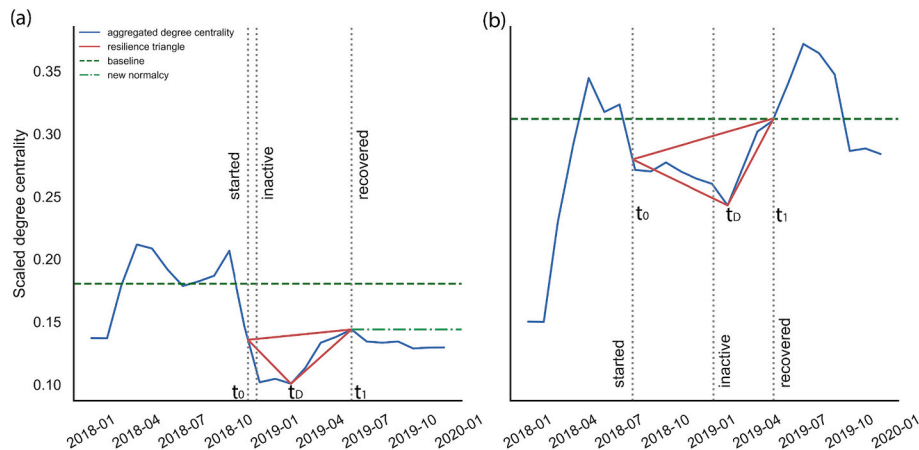


Fig. 5. Triangle detection on the average degree centrality of CBGs within the two wildfires. (a) Camp wildfire where baseline is not restored; (b) Mendocino Complex wildfire where baseline is restored.

Table 2

Description of independent variables used in the regression models.

Independent variable	Description
AvgDistance	Average distance from home CBGs to a target CBG
%PopDist<3 km	Percentage of population travel within 3 km from home CBGs to a target CBG
AreaInWildfire	The area of a target CBG within the wildfire area
# of Housing Units	Number of housing units of a target CBG
Med Household Income	Median household income of a target CBG
MedAgeMale	Median age of male of a target CBG
MedAgeFemale	Median age of female of a target CBG
# of workers	The number of full-time workers in a target CBG
%	The percentage of people that are undergraduate or higher of a target CBG
Pop>Undergraduate	

underlying covariance that impact community resilience.

3. Results and discussion

Building upon our methodology, we now turn to our findings. We will first discuss the outcomes of classified CBGs, followed by comparing the quantified resilience and two resilience metrics - vulnerability and robustness. Finally, we will investigate the potential underlying factors associated with resilience, including demographic characteristics, socio-economic status as well as built environment characteristics.

Fig. 6 demonstrates the temporal distribution of the degree centrality for every CBG within each wildfire area from January 2018 to December 2019. As the MC and Camp wildfires started on August 2018 and November 2018, respectively, we observe a sharp decline in the degree centrality after the onset of the events, indicating the immediate impacts of disruptions caused by the wildfires on human movements. Afterward, the degree centrality continues to decrease and reaches its lowest point after a period of time. Then, it gradually bounces to its pre-wildfire status or establishes a new normalcy. As we can see from Fig. 6, the lowest point of degree centrality varies among CBGs, implying that the influence of the wildfires on each CBG is different. Moreover, the time required to restore to the pre-wildfire state or establish new normalcy varies as well. Based on such different changing patterns of degree centrality, we classify the impacted CBGs into three clusters (see Fig. 6 highlighted by different colors) for each wildfire by using the DTW clustering technique discussed in Section 2.5.

In order to compare the community resilience, we detect the resilience triangle and quantify the resilience as the ratio of the area of the triangle to the total area under the pre-disruption level of performance from the initial disturbance time to the recovery time for every CBG affected by the wildfires. Fig. 7 (a) and (b) display resilience triangles for each CBG (i.e., lines with transparent color) and the overall resilience triangles for the three clusters (i.e., triangles with non-transparent color). The resilience triangle records the abrupt losses in performance of a social unit (here referring to a CBG) under the disruption of a disaster. The depth of the triangle illustrates the severity of the disruption, the length of it demonstrates the recovery time. Table 3 presents the quantified resilience of different clusters, offering a clear and direct comparison of the resilience metrics across clusters for each wildfire. As noted in Section 1, the smaller the ratio is, the higher level of resilience of the cluster. By comparing the derived resilience triangles, we find that Cluster 1 is most resilient cluster, followed by Cluster 2 and Cluster 3 in the Camp wildfire. In terms of the MC wildfire, Cluster 1 is the most resilient, followed by the Clusters 3 and 2.

To explain the potential underlying factors related to these clusters, we conduct regression analysis for each wildfire. Table 4 gives a statistical summary of the outcomes of the regression models for the two wildfires. We use the degree centrality - the proxy of resilience - as the dependent variable and create 9 factors aggregated at the CBG level as the independent variables. Through this analysis, we find for the MC

wildfire, the result shows that the model identifies around 88% of variance of the degree centrality (Adjusted $R^2 = 0.8787$) that is explained by the input variables. Further, most factors, except the number of full-time workers, percentage of people with degree higher than undergraduate, are highly correlated to the degree centrality. For the Camp wildfire, the model shows a relatively weaker ability to identify the variance of the degree centrality explained by the input variables (Adjusted $R^2 = 0.5832$). One possible reason could be that there are much more observations ($N = 950$) than that in the MC wildfire ($N = 288$), which can therefore result in more noise in the model. With respect to spatial autocorrelation of the models, no significant spatial clustering was found to be present in residuals, with Moran's I statistics being 0.1127 and 0.0983 for the Camp and MC, respectively.

Table 5 demonstrate the statistical summary of the 9 selected factors by clusters for both wildfires and Fig. B2 in Appendix displays the violin plots for each factor by clusters. In terms of MC wildfire, Cluster 1 stands out as the most resilient. It has the smallest impacted area within the wildfire and the lowest percentage of population living within 3 km of the wildfire, compared to Clusters 3 and 2 (the least resilient cluster). In addition, Cluster 1 has a relatively younger median age than other clusters and a higher percentage of education beyond undergraduate level compared to the less resilient Cluster 2, indicating a stronger demographic characteristics and socio-economic status may contribute to the resilience. In contrast, Cluster 2, the least resilient cluster, has a larger area within the wildfire and a higher population density within 3 km of the wildfire, suggesting a relatively higher level of exposure. It also has a lower socio-economic status compared to other clusters, such as lower median household income and smaller percentage of education beyond undergraduate level. However, Cluster 3, despite a larger impacted area compared to Cluster 2, shows relatively stronger socio-economic strengths with more housing units and full-time workers, which may contribute to its recovery and adaptation after the wildfire.

Regarding the Camp wildfire, Cluster 1 is identified as the most resilient cluster, followed by Clusters 2 and 3. Cluster 1 has the smallest percentage of population within 3 km of the wildfire, indicating less direct exposure to the wildfire. Second, Cluster 1 has the highest median household income and the greatest number of housing units and full-time workers, implying a stronger socio-economic foundation contributing to the resilience. Moreover, the cluster's relatively younger median age may suggest a dynamic demographic able to cope with and adapt to the disturbances caused by the wildfire. Conversely, Cluster 3, despite having the smallest impacted area within the wildfire, is quantified as the least resilient. This could be attributed to its lower number of housing units, full-time workers as well as the higher median age, which might suggest a population less able to respond effectively to the wildfire. These findings underscore the multifaceted nature of community resilience, which can be influenced by the interplay of diverse factors, ranging from the built environment to demographic and socio-economic characteristics, highlighting the necessity of considering multiple factors to gauge a community's resilience to withstand and recover from natural disasters effectively.

By leveraging the detected resilience triangles, we are able to further capture two major features of resilience - robustness and redundancy - at the CBG level (ref Section 2.4). Fig. 7 (c) and (d) show the spatial distribution of measured vulnerability and robustness, separately. The vulnerability indicates the state of being exposed to the possibility of being attacked by a disaster (i.e., wildfire in our case). The darker the color in Fig. 7 (c), the more vulnerable a CBG is. As noted in Section 1, robustness refers to the ability of a social unit to cope with the impacts of disruption after a disaster. Similarly, the darker the color in Fig. 6 (d), the stronger robustness a CBG is.

4. Conclusion

As noted in introduction (Section 1) of the paper, disasters can

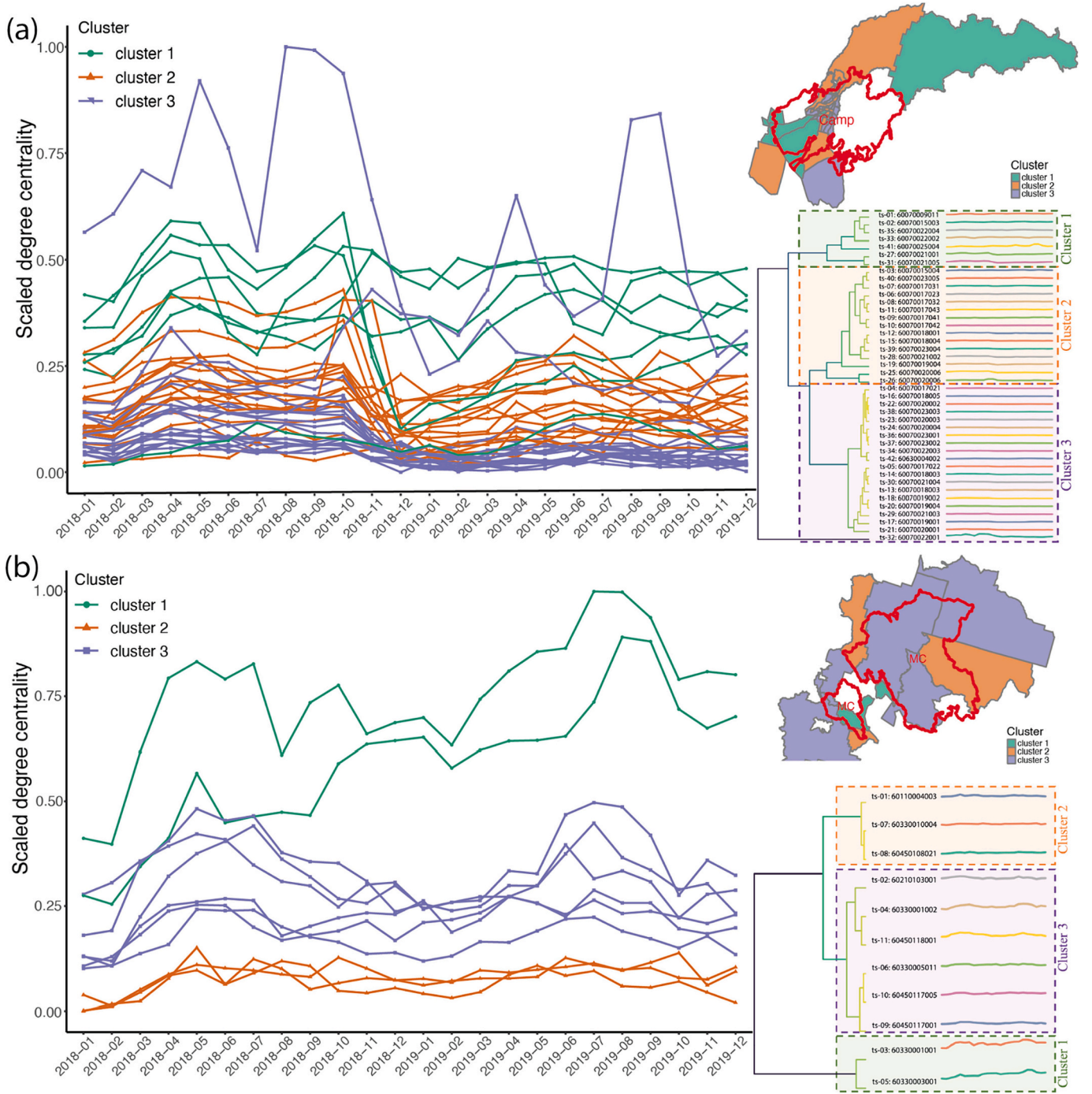


Fig. 6. The distribution of degree centrality for each census block group colored by different clusters. (a) The Camp wildfire; (b) The Mendocino Complex wildfire. (Note: the degree centrality is normalized by using min-max normalization to ease comparison.)

severely impact societies at large and to quantify community resilience to such events remains an open research question. In this paper, we proposed a framework to quantitatively measure and compare the disaster resilience of communities based on human mobility data collected from SafeGraph over space and time. We do so by utilizing network analysis to measure the degree centrality as the proxy of resilience as it reflects the exposure of a community to a disaster and their response to it. The measured degree centrality is then utilized to derive the resilience triangle of a community, which is used to quantify and compare resilience among communities. By leveraging DTW clustering and regression analysis techniques, we are able to classify similar communities into clusters and uncover underlying factors that affect the

resilience of the given community, such as demographic, socio-economic as well as built environment characteristics.

Using the Mendocino Complex and Camp wildfires in California as test cases, and human mobility data in and around these fires, we illustrated how the proposed methodology can be utilized to capture disaster resilience at the CBG level from January 2018 to December 2019. By comparing the dynamic changes of degree centrality and the derived resilience triangles during the study period, we found that the wildfires drove a sharp decline in degree centrality, implying an immediate impact of these disruptions on human movements. However, the level of the disruption and the span of time needed to bounce to pre-wildfire levels or to establish a new normalcy varied among CBGs. In

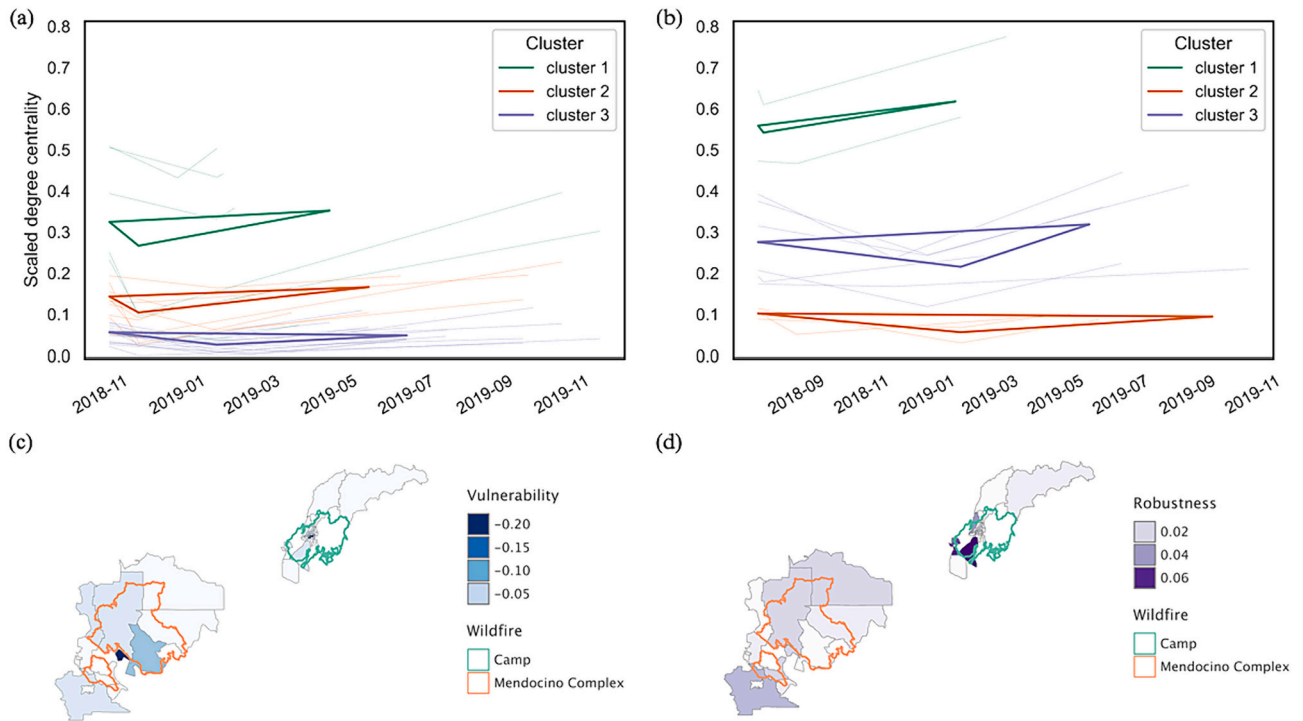


Fig. 7. The results of resilience triangles of clustered CBGs and resilience features. (a) The determined resilience triangles of clustered CBGs for Camp wildfire; (b) The determined resilience triangles of clustered CBGs for Mendocino Complex wildfire; (c) Vulnerability of CBGs within the two wildfire areas; (d) Robustness of CBGs within the two wildfires.

Table 3
Quantified loss of resilience and resilience of different clusters for each wildfire.

	Camp			MC		
	Cluster 1	Cluster 2	Cluster 3	Cluster 1	Cluster 2	Cluster 3
Loss of resilience (i.e., the area of the triangle)	0.007	0.006	0.004	0.002	0.012	0.018
Resilience (i.e., the ratio of the triangle to the total area under the pre-disruption level of performance)	9.10%	13.40%	26.14%	1.56%	21.52%	14.57%

Table 4
Statistical summary of regression models for the two wildfires.

	Dependent variable: Degree Centrality	
	Mendocino complex	Camp
(Intercept)	7.8055*** (0.4516)	6.2647*** (0.1613)
AvgDistance	0.0000* (0.0000)	− 0.0000 (0.0000)
%PopDist < 3 km	− 0.0137*** (0.0024)	0.0104*** (0.0009)
%AreaInWildfire	− 0.0000** (0.0000)	0.0000*** (0.0000)
# of Housing Units	0.0017*** (0.0005)	0.0022*** (0.0002)
Med Household Income	0.0000 (0.0000)	0.0000*** (0.0000)
MedAgeMale	− 0.0013 (0.0025)	− 0.0060** (0.0019)
MedAgeFemale	− 0.0311** (0.0094)	− 0.0125*** (0.0021)
# of workers	− 0.0019 (0.0012)	− 0.0025*** (0.0003)
% Pop > Undergraduate	0.0259 (0.0210)	− 0.0239*** (0.0043)
Cluster 2	− 1.3039*** (0.0928)	− 0.6918*** (0.0653)
Cluster 3	− 0.6181*** (0.0785)	− 1.1131*** (0.0674)
R ²	0.8837	0.5884
Adj. R ²	0.8787	0.5832
Num. obs.	288	950

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

addition, our results also revealed that both socio-economic status and demographic characteristics play a crucial role in assessing resilience. These findings confirm the multifaceted nature of community resilience, which highlights the necessity of considering multiple factors to gauge a community's ability to withstand and recover from the disturbances of natural disasters effectively.

The study presented here provides only an illustrative case study of the proposed framework and can certainly be expanded. First, we only tested the methodology on SafeGraph data, which is a type of human mobility data, while other human mobility data, such as social media, could theoretically be compatible with it. Thus, additional research is needed to examine the replicability of our method on different data. Secondly, the SafeGraph data may have certain limitations, such as biases at an aggregation level (e.g., state, county, tract, CBG) and in demographics (e.g., age, race, household income, education) (SafeGraph, 2019). However, we consider this open research question beyond the scope of our current paper as we focus on quantifying resiliency based on mobility data. As such, future studies could explore efforts to generalize these types of data to overcome any such biases before being used for policy making (Coston et al., 2021). Moreover, apart from a quantitatively interpretation of individual CBG results, the regression analysis provides only an initial exploration of the potential covariance that impacts community resilience. Furthermore, the assumption of using relatively time-constant variables (i.e., census data) to explore the relationship to the mobility data may miss the dynamic environments

Table 5

The statistical summary of different factors for Mendocino Complex and Camp.

Cluster 1	N		Mean		St. Dev.		Min		Max	
	MC	Camp	MC	Camp	MC	Camp	MC	Camp	MC	Camp
Average distance (km)	48	168	108.293	59.584	23.705	25.164	49.634	15.132	157.065	176.266
% Population within 3 km	48	168	38.454	33.528	6.088	13.510	24.054	6.993	48.921	63.515
Area within wildfire (km ²)	48	168	27.083	16.885	26.847	24.727	0.517	0.791	53.649	71.604
# of housing units	48	168	634.500	700.143	63.161	188.709	572	437	697	1099
Median household income	48	120	50.130	55.666	4.493	18.078	45.684	38.917	54.575	90.156
Median age (Male)	48	168	37.700	43.457	3.436	3.545	34.300	38.400	41.100	48.800
Median age (Female)	48	168	39.450	45.400	0.152	7.748	39.300	33.900	39.600	59.100
# of full-time workers	48	168	347.000	364.857	2.021	233.517	345	113	349	835
% Education >Undergraduate	48	168	4.393	6.444	1.848	2.902	2.564	2.291	6.222	9.288
Cluster 2										
	MC	Camp	MC	Camp	MC	Camp	MC	Camp	MC	Camp
Average distance (km)	72	360	80.153	54.779	33.502	21.108	22.420	10.300	185.619	153.289
% Population within 3 km	72	360	44.528	46.371	12.382	20.022	16.406	2.469	64.286	80.000
Area within wildfire (km ²)	72	360	122.393	5.617	124.615	8.359	11.132	0.006	295.008	30.579
# of Housing units	72	360	372.000	543.067	14.823	227.980	357	303	392	1037
Median household income	72	360	60.166	48.675	26.471	13.873	38.542	24.565	97.165	73.194
Median age (Male)	72	360	45.333	44.287	9.850	9.533	32.900	26.500	56.800	58.800
Median age (Female)	72	360	44.033	45.393	10.362	11.226	31.300	24.900	56.500	59.400
# of full-time workers	72	360	196.667	287.133	120.218	156.896	78	107	360	715
% Education >Undergraduate	72	360	3.455	6.947	1.938	4.749	0.733	0.000	4.843	16.667
Cluster 3										
	MC	Camp	MC	Camp	MC	Camp	MC	Camp	MC	Camp
Average distance (km)	168	470	89.281	50.112	40.419	24.681	23.355	6.355	202.424	152.443
% Population within 3 km	168	470	45.462	39.990	14.947	22.724	16.354	2.010	72.704	91.209
Area within wildfire (km ²)	168	470	178.212	2.934	244.604	2.561	0.018	0.745	702.816	11.928
# of housing units	168	470	533.429	473.370	243.628	165.208	248	170	1034	928
Median household income	168	470	58.407	55.593	20.280	16.947	26.250	25.469	81.875	96.844
Median age (Male)	168	470	44.386	50.734	10.077	12.270	26.800	20.700	56.900	64.300
Median age (Female)	168	470	45.771	52.907	7.523	11.744	34.900	27.700	69.100	53.600
# of Full-time Workers	168	470	209.429	226.309	49.965	65.624	117	87	259	360
% Education >Undergraduate	168	470	6.524	6.495	3.563	4.855	1.370	0.000	11.711	15.909

caused by wildfire events. This is one reason why we share the source code and relevant aggregated data to allow other researchers to apply our method to their own areas of research and/or to extend our work. Building on what has been discussed above, further research could also explore more complex styles of models (e.g., Bayesian hierarchical models) to obtain a more in-depth understanding of the causal processes behind the resilience and impacts of wildfire disruption on human movements. In addition to these considerations, it is crucial to note that the direct impact of wildfire events on human mobility patterns is complex due to the potential interplay of multiple factors such as economic conditions and social dynamics. Therefore, future studies could focus on isolating the specific impact of wildfires from other confounding variables, thereby deepening our understanding of resilience quantification.

Nonetheless, this research further widens the use of mobility data in the context of disaster resilience, especially for understanding disaster and their prolonged impacts on society. It opens the door to more informed evaluations of resiliency of a community to the impacts of a disaster, which can be used to make informed decisions for future emergency management and policy making.

Appendix A. Examples of SafeGraph data used in this study

CBG GEOID	Date start	Date end	Device home CBG GEOID	Device count
060375760011	2018-01-01	2018-02-01	060375033012	4
060375760011	2018-01-01	2018-02-01	060790118001	4
060971501002	2018-01-01	2018-02-01	060552013001	5
060971501002	2018-01-01	2018-02-01	060971501002	201
060710104021	2018-01-01	2018-02-01	060710104021	1348

Author statement

Qingqing Chen: Methodology, Software, Formal analysis, Visualization, Writing - Original Draft, Review & Editing.

Boyu Wang: Methodology, Formal analysis and Visualization.

Andrew Crooks: Supervision, Conceptualization, Methodology, Writing - Review & Editing.

CRediT authorship contribution statement

Qingqing Chen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Boyu Wang:** Visualization, Methodology, Formal analysis. **Andrew Crooks:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

None.

Appendix B. Figs

- B1. An example of degree centrality sequence alignment of two CBGs.
 B2. The comparison of different factors across clusters. (a) The Mendocino Complex wildfire; (b) The Camp wildfire.

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